

Acute Procedural Pain in Children Intervention With the Hospital Clown

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Introduction: Hospitalized children often describe needle-related procedures as the worst pain possible and such procedures may be emotionally traumatic. The use of hospital clowns (HCs) related to painful medical procedures in children may offer pain relief, but this has not been systematically evaluated. The objective of this study was to assess the effect of a therapeutic clown in comparison with standard care on the experience of pain in children receiving venipuncture.

Materials and Methods: A sample of 116 children aged 4 to 15 years consecutively admitted to the hospital was allocated to either the experimental (presence of HC) or control group (standard care) before venipuncture. Self-reported pain after the procedure was assessed using the Faces Pain Scale combined with a 0 to 10 Numerical Rating Scale. Separate analysis was conducted in age groups from 4 to 6 (N=37) and 7 to 15 (N=74) years.

Results: Without the clown present, the mean pain score (2.7 ± 2.8) was not significantly different between the 2 age groups. Children aged 7 to 15 years had lower pain scores when the clown was present compared with the control group ($P=0.025$). Children aged 4 to 6 years had higher pain scores with the clown present, although the difference was not statistically significant ($P=0.054$). Children with pain (N=49) or previous experiences with venipuncture (N=56) did not score pain significantly differently.

Discussion: Assessing the pain experience of children receiving venipuncture with the presence of an HC indicates a pain relieving effect for children older than 6 years. However, future studies should carefully study the effects on younger children where mixed effects may be present.

Key Words: hospital clown intervention, venipuncture/procedural pain, pain assessment, self-report, children

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Hospitalized children experience fear and anxiety because of separation from their family and friends, the unfamiliar environment, and potential painful procedures and treatment.¹ Children often describe needle-related procedures as the most painful aspect of the health care experience.^{2–4} Studies demonstrate that a high frequency of unmanaged pain puts hospitalized children of all ages at risk for both immediate and long-term physiological and behavioral consequences.^{2,5–9}

In cases of acute hospitalization, time to prepare for painful procedures is limited. As a result, the experience is not only more painful for children, but it can cause distress to parents.^{10,11} Painful procedures may not only be immediately traumatic. Indeed, children experiencing untreated procedural pain and children with negative previous procedural experiences have an increased risk of anxiety and distress, with psychological consequences for subsequent procedures and future health care behaviors.^{9,12,13} Thus, a focus on management of pain is essential in the context of an acute painful procedure.¹⁴ To meet this need, the International Guideline in Pediatric Anesthesia (Good practice in postoperative and procedural pain management)¹⁰ recommends pain assessment as well as pharmacological and nonpharmacological strategies to prevent and effectively manage acute procedural pain in children.

Pharmacological strategies for intravenous cannulations involve the usage of topical local anesthetics.^{10,15} However, studies indicate that topical local anesthetics do not guarantee pain relief, and children with anxiety may not be helped by this intervention. Thus, there is a need to reduce anticipatory and procedural anxiety using non-pharmacological strategies.^{2,10,11,16} Nonpharmacological strategies attempt to shift the child's focus to something more attractive and draw attention away from pain stimuli.^{7,15,17–19} Stinson et al⁹ reported that acute procedural pain in children was efficiently reduced by distraction and hypnosis. Uman et al¹⁹ had similar results, highlighting the effectiveness of distraction, hypnosis, and combined cognitive-behavioral interventions to reduce the pain and distress that accompany these procedures. These findings are in line with other studies focused on the impact of non-pharmacological strategies that involve “moving the focus away from the painful situation” and “guiding the child through the procedure.”^{14,20}

A hospital clown (HC) is 1 example of a non-pharmacological distraction strategy.²¹ The HC is a professional with a specific practical and theoretical education who uses cognitive distraction and imagery to shift attention away from pain or sources of distress and thereby encourages the child to cope.^{22–25} In the last decade, there has been a rapid growth in the presence of HCs in pediatric hospitals

as part of the interdisciplinary teams present around the child and family. The HC is still mostly used in the context of entertainment, on the basis of the assumption that humor is associated with increased well-being of the child.^{21,26,27} Although many pediatric units routinely use them, HCs are rarely asked to interact with children during painful procedures.²² Only a few randomized studies have investigated the usage of HCs related to painful procedures. Felluga et al²⁸ reported no beneficial effects on pain levels in children aged 4 to 11 years undergoing 5 different therapeutic procedures. These results are consistent with Meiri et al²⁹ reporting no benefit on pain level in children aged 2 to 10 years undergoing venipuncture or intravenous cannulation in comparison with a control group, using topical anesthetics (EMLA). Conversely, their study showed a beneficial effect on anxiety and length of crying with the HC. Another study including children aged 3 to 16 years undergoing either venipuncture or intravenous cannulation found a pain reduction in 3- to 7-year-old children with the presence of an HC, although there was no change in 8- to 16-year olds.²⁵ However, research on the effects of systematic interventions with an HC on children's pain experience undergoing a standardized procedure is limited. Furthermore, major questions remain as to how different age groups react to the presence of a clown as an intervention and whether the potential effect has any relation to previous experiences of pain.

This study aimed to evaluate the effect of an intervention with the HC on self-reported pain level compared with standard care (SC) for children receiving venipuncture in the pediatric admission unit at a hospital. The study specifically focused on examining potential differences in intervention effect on the basis of (1) age group (4 to 6 vs. 7 to 15 y) and (2) presence of a pain condition before venipuncture or previous pain experiences.

MATERIALS AND METHODS

Study Design and Material

This study is a prospective, nonblinded trial. A parallel design was used that studied children aged 4 to 15 years who were admitted consecutively at a Danish University Hospital over a period of 9 months, from April 2016 to January 2017. They were assigned to a HC intervention or SC. Children were considered eligible for the study if they were (1) acutely admitted to the hospital at the admission unit and scheduled for venipuncture procedure, (2) not prepared for the procedure before coming to the hospital (not knowing about having a venipuncture), and (3) accompanied by at least 1 parent. Children with developmental disorders who could not cooperate with self-reporting of pain or who were unable to speak Danish or English were excluded. The study was approved by the pediatric administration at the University Hospital. Permission was given by the Data Inspectorate (id: 2016-5). The local ethics committee was consulted, and they certified that the study did not require their approval.

Written and oral information were provided to both children and parents. Written informed consent for study participation was obtained from parents.

With expected deviations of pain scores of 10% to 20% (estimation for clinical relevance),^{30–32} a sample size of 52 children in each group was estimated, assuming α -level of 5% and 80% power.

Allocation

There were specific time frames planned for the study (requiring presence of the HC). The HC was available on certain days and was scheduled for all day on weekdays and for some evening hours during the study period. This schedule was established for the entire inclusion period before starting child recruitment. When the HC was available, the HC took part in all the venipuncture procedures for the eligible children. Children were considered as assigned to the HC group if they were admitted on days when the HC was available and to the SC group otherwise. Pilot data collection 1 month before the study was conducted to investigate the distribution of age, sex, and diagnoses over weekdays. This pilot confirmed that, of 173 diagnoses, sex and age were uniformly distributed on all days of the week and hence independent of the presence of the HC. In summary, eligible children were recruited during all weekdays independently of the presence of the HC. Hence, their assignment was considered random, and the data were analyzed accordingly. In the SC group, 10 children were included in the summer period, during which the HC was absent for 5 weeks.

Intervention

Before this study, the HC was not part of the daily routine and practice at the participating unit. Children in the HC group met the clown before venipuncture procedures either in the waiting room or in the examining room. One male HC was used in the study in order to ensure minimal variation in technique. In the HC group, the children and their parents interacted with the HC while waiting for the procedure and during the time needed to complete the venipuncture. Depending on the child's age, emotional state, and cognitive development, the HC used different methods. Overall, the performance of the HC included creating a relationship with the child by using different techniques.

The HC in this study utilized different techniques by building a relationship with the child before the venipuncture, specifically by using music, songs, toys, and making agreements for the procedure (eg, looking or not looking at the needle) in collaboration with the child, parents, and health care personnel. During the procedure, the HC had a conscious use of distraction techniques such as holding the child's hand, temporary tattoos (a small sticker/label with a picture applied to the skin with water), dream journeys, songs, and storytelling.

Children in the SC group were exposed to the other clinical staff's individual distraction, comfort, and care techniques for acutely hospitalized children undergoing painful medical procedures in collaboration with the parents. In the SC group, nurses interacted with the child from the beginning of admission and prepared the child by explaining verbally the venipuncture procedure. Under the procedure, the biomedical laboratory technologist guided and structured the procedure.

Different pediatric nurses and biomedical laboratory technologists assisted with the procedures, and they were randomly distributed in both groups. In both groups of children, the parents, the nurses, or the biomedical laboratory technologists were not prepared to use any specific instructions on how to aid the child through the procedure. For both groups, the procedure was performed in 2 examining rooms of equal size and with the same interior arrangements. The walls were decorated with colorful paintings and characters suitable for children. The procedure was performed on a chair or on the bed.

Data Collection Procedures

Baseline information including age, sex, presence of a current painful condition, and previous experiences with venipunctures was obtained in the examining room.

The primary outcome measure was the child's self-report of pain on an age-appropriate scale. The self-reporting of pain was carried out in the examining room immediately after finishing the venipuncture, during a time interval of between 0 and 5 minutes. No studies were found that investigated the specified time interval for self-report of pain after a venipuncture. Thus, this time interval was chosen on the basis of a pilot study, which was carried out in the pediatric setting on 11 randomly selected children in February 2016. This time interval was validated with ethnographic field notes. No systematic difference was found between the 2 groups in the pilot study.

In addition, the number of punctures for each child and duration of each procedure was documented because of their potential relation to the exposure.

Pain Assessment

Self-reported pain assessment is recommended as the primary source for information about pain intensity, accounting for differences in age as well as cognitive and communicative abilities. Most children have developed the ability to provide an accurate self-report from 3 years of age.^{33,34}

Thus, the Numerical Rating Scale (NRS) combined with the Wong-Baker Faces Pain Scale (FPS) was chosen for pain assessment. These 2 scales are widely used in clinical pediatric practice.³⁵ The NRS is recommended for children aged 8 years and older; the FPS is recommended

for ages 4 to 12 years.^{10,11,32,34} The NRS ranges from 0 on the left side, indicating "no pain," to 10 on the right side, indicating the "worst pain possible." The FPS is a horizontal scale of 6 drawn faces that range from a smiling "no hurt" anchor face on the left side to a crying "hurts worst" anchor on the right. In this study, the FPS was coded to a 0 to 10 scale.^{34,36} The 2 scales were depicted on the same laminated paper, with parallel presentation of the numbers and the 6 faces. This version was chosen to secure a developmentally, cognitively, and culturally appropriate pain scale for the age group in the present study (4 to 15 y) as well as to secure a version easy to use in clinical practice.^{12,32} While age is a proxy for developmental level, chronological age is not the sole indicator of a child's capacity to self-report pain. Different children have different assessment needs^{10,11,35}; thus, independent of age, children were asked to choose either the numbers or the faces on the scale, substantiated by their different cognitive capacities and developmental level. Both scales were well known and routinely used among the nurses in the unit.

Statistics

Statistical analyses were conducted using STATA version 14 software. Fisher exact test was used to assess potential differences between the 2 groups in categorical characteristics of participants. An unadjusted unpaired 2-sample *t* test was conducted to compare the pain scores and other respective characteristics of the 2 groups. Adjusted analysis by multivariable linear regression was performed to capture the effect of potential intermediate variables (local anesthetic, nitrous oxide, time per procedure, number of punctures).³⁷

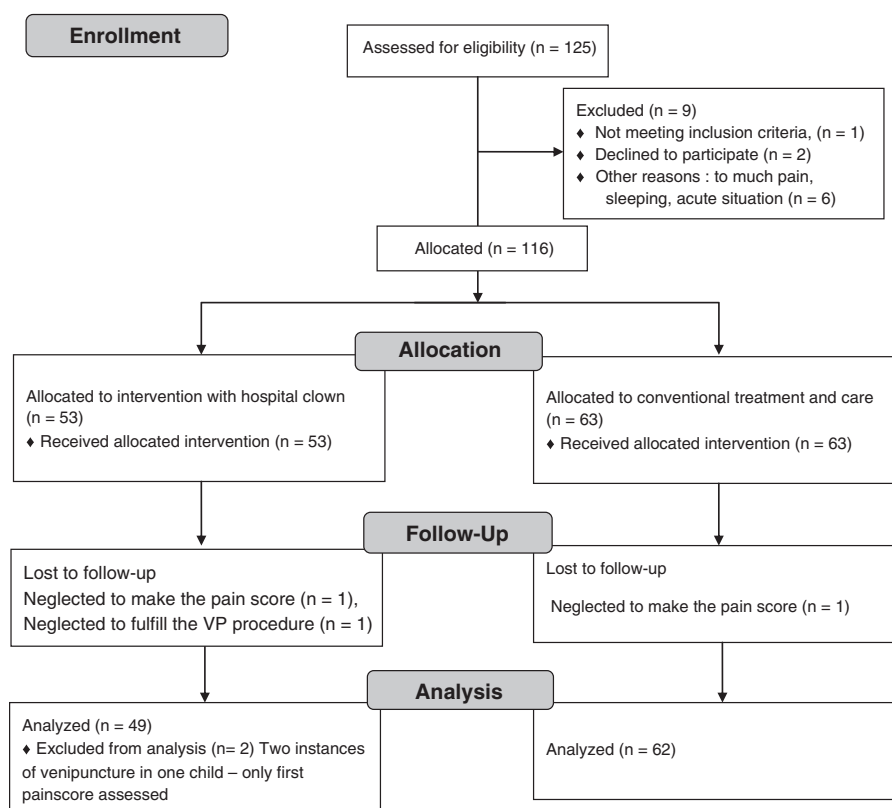


FIGURE 1. CONSORT flow diagram of sampling.

Supplementary analyses were performed, stratified by age group (4 to 6 and 7 to 15 y), sex, pain condition, and previous venipunctures. By multivariable linear regression, the effects among these strata were compared. In all analyses, the eventual non-normality and variance of in-homogeneity were addressed using bootstrapping with 1000 replications to estimate standard errors.

The cut off for age groups (4 to 6 and 7 to 15 y) was based on the literature addressing challenges in providing self-report of pain in younger preschool children.^{33,38}

Sensitivity analyses with respect to population heterogeneity because of the summer period, use of nitrous oxide, and 14 to 15 years of age was carried out by omitting the few relevant cases from the analyses. Significant findings were found for P -values ≤ 0.05 .

RESULTS

A total of 116 children were enrolled in the study. After allocation to either the HC or the SC group, 1 in each group refused to make the self-report ($n=2$). Furthermore, 1 child refused to fulfill the venipuncture, and 2 children had 2 venipuncture procedures in the same session; thus, only the first score was included in the analysis. In total, 111 children were included in the analysis: 49 in the HC group and 62 in the SC group (Fig. 1).

No demographic differences between the children assigned to the HC group ($n=49$) or the SC group ($n=62$) were found (Table 1). When comparing the use of local anesthetics, nitrous oxide, number of punctures, and

duration of procedure, no significant difference between the 2 groups was found.

The analysis of the entire group (4 to 15 y) found no statistically significant differences in pain scores between the 2 groups with or without adjusting for potentially influencing variables (Table 2).

The stratified analysis showed an effect of the HC's presence varying between age groups. Pain scores in the 7 to 15 years group showed a positive effect of the presence of the HC, with a decrease in the scores (Table 2). When adjusting for the effect of potential intermediate variables, similar effects were identified ($P=0.025$).

Conversely, in the group aged 4 to 6 years, we found a higher pain score with the presence of the HC (Table 2), although this difference was not statistically significant ($P=0.087$). When adjusting for the effect of potential intermediate variables, the difference was enhanced but still not statistically significant ($P=0.054$).

A statistically significant difference in effect of the HC between the age groups (4 to 6 and 7 to 15 y) was found ($P=0.018$) (Table 2). By effect, we mean the difference in score between the exposed and the unexposed children.

There were no differences related to sex found between the 2 groups ($P=0.997$).

From the stratified analyses, no significant difference was found for children affected by the presence of a painful condition and children with previous venipunctures, with or without the presence of the HC (Table 2). However, the estimated effect for children with no previous venipunctures $[-1.2 (-2.5, 0.2), P=0.085]$ provides some support for a benefit of the presence of the HC. Conversely, for children with previous venipunctures, the estimated effect was negative $[1.0 (-0.7, 2.6), P=0.248]$.

There was no statistically significant difference between the 2 groups when comparing the presence of a painful condition/no presence of a painful condition $[1.4 (-1.0, 3.8), P=0.254]$, whereas with respect to previous venipunctures, there was a difference $[2.1 (-0.0, 4.3), P=0.054]$ (Table 2).

DISCUSSION

For the entire sample (4 to 15 y), no statistically significant difference in pain scores between the SC and HC groups was found. The literature scarcely reports on the question of HCs' effect on children undergoing venipunctures. Nonetheless, the results of the present study are to some extent in accord with a few other studies assessing pain related to painful procedures with HC intervention.^{25,28,29} Felluga et al²⁸ and Meiri et al²⁹ found no significant pain reduction with the intervention of an HC but did find a positive effect on anxiety in children aged 4 to 11 and 2 to 10 years, respectively. In contrast to the findings from our study, Wolyniez et al²⁵ found a tendency for the pain score to be lower with the presence of an HC in children younger than 7 years and unchanged in the older age group (8 to 16 y). However, these studies have not investigated homogenous painful procedures and have included different age groups and contexts. Thus, the results of our study should be interpreted within the specific context, procedure, and age groups.

The overall results may partly be explained by the fact that anxiety and fear may affect the pain level independent of the presence of the HC. The HC intervention can be seen as a nonpharmacological strategy similar to distraction, which seems effective for pain relief associated with painful procedures.^{7,18,19} Distraction is defined as a cognitive or

TABLE 1. Homogeneity in General Characteristics of Participants (N = 111)

Characteristics	n (%)		P
	Hospital Clown (N = 49)	Standard Care (N = 62)	
Sex			
Boy	27 (55.1)	36 (58.1)	0.848
Girl	22 (44.9)	26 (41.9)	
Age (y)			
4-6	15 (40.5)	22 (59.5)	0.686
7-15	34 (46.0)	40 (54.0)	
Age (mean $\pm$ SD) (y)	9.1 $\pm$ 3.4	8.6 $\pm$ 3.4	0.396
Local anesthetic			
Yes	32 (65.3)	38 (61.3)	0.396
No	17 (34.7)	24 (38.7)	
Nitrous oxide			
Yes	2 (4.1)	1 (1.6)	0.582
No	47 (95.9)	61 (98.4)	
Previous venipuncture			
Yes	27 (55.1)	29 (46.8)	0.446
No	22 (44.9)	33 (53.2)	
Pain			
Yes	18 (36.7)	31 (50.0)	0.182
No	31 (63.3)	31 (50.0)	
Duration (mean $\pm$ SD) (min)	5.2 $\pm$ 2.4	5.7 $\pm$ 3.8	0.316
No. punctures			
1	44 (89.8)	56 (90.3)	1.000
> 1	5 (10.2)	6 (9.7)	
Pain score			
NRS	14 (28.6)	26 (41.9)	0.167
FPS	35 (71.4)	36 (58.1)	

FPS indicates Faces Pain Scale; NRS, Numerical Rating Scale.

TABLE 2. Crude and Adjusted* Analysis

Variables	Control Group (Mean ± SD)	Hospital Clown (Mean ± SD)	Crude Difference (SE/CI)	P	Adjusted Difference* (SE/CI)	P
Total	2.7 ± 2.8	2.5 ± 3.2	−0.1 (−1.3, 1.0)	0.819	−0.1 (−1.2, 1.0)	0.920
Age (y)						
4-6	2.6 ± 3.8	4.9 ± 4.1	2.2 (−0.3, 4.8)	0.087	2.6 (−0.1, 5.2)	0.054
7-15	2.7 ± 2.0	1.5 ± 2.0	−1.2 (−2.1, −0.3)	0.010	−1.0 (−1.9, −0.1)	0.025
Diff. (4-6/7-15)			3.4 (0.6, 6.8)	0.016	3.6 (0.6, 6.5)	0.018
Sex						
Boys	2.4 ± 2.5	2.2 ± 2.9	−0.2 (−1.5, 1.2)	0.837	−0.03 (−1.4, 1.3)	0.960
Girls	3.0 ± 3.1	2.9 ± 3.5	−0.2 (−2.0, 1.7)	0.867	−0.3 (−2.0, 1.5)	0.754
Diff. (boys/girls)			0.0 (−2.3, 2.3)	0.997	0.2 (−2.1, 2.6)	0.838
Pain						
Yes	3.0 ± 3.2	3.9 ± 3.7	0.9 (−1.1, 2.8)	0.389	0.8 (−1.3, 2.8)	0.466
No	2.3 ± 2.3	1.7 ± 2.6	−0.6 (−1.8, 0.7)	0.371	−0.4 (−1.6, 0.8)	0.535
Diff. (yes/no)			1.4 (−1.0, 3.8)	0.254	1.1 (−1.1, 3.4)	0.332
Previous venipunctures						
Yes	1.9 ± 2.4	2.8 ± 3.8	1.0 (−0.7, 2.6)	0.248	1.1 (−0.4, 2.6)	0.163
No	3.4 ± 2.9	2.2 ± 2.2	−1.2 (−2.5, 0.2)	0.085	−0.9 (−2.4, 0.5)	0.202
Diff. (yes/no)			2.1 (−0.0, 4.3)	0.054	2.0 (−0.0, 4.0)	0.053

*Adjusted for baseline variables related to the procedure: local anesthetic, nitrous oxide, time per procedure, number of punctures.
CI indicates confidence interval; Diff., difference.

behavioral strategy that redirects the child's attention or actively involves the child.¹⁸ The HC strategy differs from other distraction by tailoring the strategy to each individual child choice and preferences.¹⁷

However, a strong correlation exists between perceived pain and affective and cognitive aspects of the pain experience.³² Anxiety, fear, and distress can exacerbate children's pain experience.²⁵ Thus, it is preferable to use self-reporting in combination with observational measures of behavior in clinical practice with special attention on younger children.^{32,34}

Overall, the preferred pain scoring method in both groups was the FPS (Table 1), which is consistent with earlier reviews on self-reporting of pain.^{34,35} Previous studies on pain assessment in younger children have shown challenges in using scales, as the children may lack the requisite comprehension level to use measures and to some extent favor the extreme ends of scales.^{35,39} However, the results of the present study did not show a significantly higher score in the 4- to 6-year-old SC group (2.6 ± 3.8) compared with the score in the 7- to 15-year-old SC group (2.7 ± 2.0), indicating no favoring of the high end of the scale in the 4- to 6-year group. Thus, these overall results raise questions with regard to HC intervention for different age groups.

When conducting the stratified analysis of the data divided into 2 age groups, an increase in pain score was found in the 4- to 6-year-old group with the presence of the HC, although the difference was not statistically significant ($P=0.054$). In contrast, a decrease was found in the 7- to 15-year-old group with the presence of the HC. Adjusting for intermediate variables, the results showed similar effects, meaning that the observed difference in pain scores cannot be explained by the intermediate variables.

These age-related differences are in contrast with Wolyniez et al,²⁵ who found improved pain scores in younger children, whereas in older children, pain scores were unchanged. The results are to some degree consistent with Hansen et al,²⁶ who found sex-specific differences, with a negative effect on boys younger than 8 years.

Thus, the negative effect on younger children in this study may have several different explanations. First, compared with older children, younger children may be especially affected by cognitive, emotional, or situational factors³⁵; such factors may affect the child's collaboration with the HC and thus explain the increase in pain scores in the 4- to 6-year-old group.

Second, the HC represents an additional and unknown person in a challenging situation. Even when the HC met the child and family before the procedure, there may have been some challenges in making an alliance in a situation typically characterized by fear, pain, and distress. Although no fear of clowns was registered, Linge^{23,24} describe age-related differences in the alliance, pointing out that younger children meet the HC with more caution; as a result, an alliance requires time to develop. Therefore, creating the alliance in an acute situation may be a challenge. Third, the negative effect on children aged 4 to 6 years may partly be explained by the fact that younger children have a lower perceived level of control to allow them to benefit from a distraction.¹⁴ Although the results of the present study did not show a higher score in the 4- to 6-year-old SC group (2.6 ± 3.8) compared with the score in the 7- to 15-year-old SC group (2.7 ± 2.0), the majority of research on rating scales indicate that younger children might experience difficulties using the middle points of scales when rating emotional states.^{39,40} Their limited cognitive abilities with regard to complex abstract thoughts, compared with older children with a more abstract thinking and logic must be taken into account, when interpreting the results from this study, in particular the results in children aged 4 to 6 years. Further research is required within this group.

The results revealed by this study showed a non-significant difference between the 2 groups related to prior venipunctures ($P=0.054$). The estimated effect for children with no previous venipunctures (−1.2) provides some support for a benefit of the presence of the HC that is large enough to be considered clinically important. Although this did not reach statistical significance ($P=0.085$), the confidence

interval leaves open the possibility of a positive effect (-2.5) of the HC intervention. This is large enough to be clinically important, and it provides strong evidence against a substantial negative effect (0.2). Conversely, for children with previous venipunctures, the estimated effect was negative (1.0). Thus, the confidence interval argues against a substantial positive effect (-0.7) but leaves open the possibility of a substantial negative effect (2.6) of the HC intervention. This result supports that earlier experiences (and thus level of distress) are a major factor in children's ability to cope. Earlier negative painful experiences can lead to a current higher pain score.^{12,26} Comprehensive research on psychological sequelae after unmitigated pain concerning vaccination of children point out the adverse impact these memories may have on fear and pain at future procedures.¹³ However, in our study, we did not ask whether the earlier experience was either negative or positive. Taking into account that children often report pain related to needles as "the worst pain possible,"²⁻⁴ it may be a plausible explanation that memories of previous painful experiences are characterized by children as negative and therefore seem to have great influence on the child's expectancy of pain, level of distress, and efficacy of pain strategies.^{2,14} Thus, children with previous experiences benefit, to a minor degree, from the intervention.

Strengths and Limitations

The results of this study should be considered in the light of the following limitations. First, the results may be somewhat limited by the small sample size, when divided into 2 age groups (Fig. 1). Therefore, the study only demonstrated a tendency in increases and decreases in pain level, respectively. Further research should be undertaken in different age groups with larger samples.

Second, we used self-reported pain as the only pain measurement. Self-reporting of pain reflects different perspectives³²; thus, perfectly reliable and valid measurement is unattainable.³⁴ Nonetheless, self-reporting of pain is considered as the preferred approach.¹⁰

Although children's self-reporting of pain intensity is a valuable source of information on pain, care must be taken in interpreting pain intensity from standardized pain scores alone. To some extent, we have taken this into consideration by choosing different age-appropriate scales and letting the child choose their preferred scale. Third, in this study, a pain measurement tool was used that combined the FPS with NRS. This approach could have resulted in response bias, as younger children tend to choose the FPC and have a tendency to use the extremes of the scale.³⁵ However, all children were instructed in both scales, and registration showed a random distribution of choice of scale in both HC and SC groups (Table 1). Furthermore, the results from the SC group for both age groups illustrated no clinically significant differences. Thus, we suggest that there is no response bias with regard to this issue in the present study. Despite this finding, it is important to point out that the presence of the HC and other predictors can only explain some of the variances in the children's scores. Thus, these results might also be explained by other factors (eg, the affective impact, children's control and coping, parental presence and attitude) not taken into account in this study. The parent's role and perspective might be addressed in further studies as well as the nurses' and HC's perspective.

The allocation of children to either the experimental (presence of HC) or control group (SC) depends on the availability of the HC. This limitation is handled by conducting a pilot study before the current study. Results from

the pilot showed a random distribution of children's diagnosis, age, and sex on all days of week, and thereby strengthens the allocation.

A strength of this study was its use of only 1 HC, validating a uniform HC intervention for all included children. Another strength is that a homogeneous painful procedure, venipuncture, was investigated. A final strength was that usage of the NRS and FPS, depicted on the same laminated paper, was easy, quick, and familiar to all the pediatric nurses in the unit.

CONCLUSIONS

The present study contributes to the existing evidence with regard to nonpharmacological strategies for acute pain management. Specifically, the study adds new insights with regard to the effect of the HC as an individual tailored strategy, and addresses an understudied topic for a nonpharmacological intervention associated with venipunctures in acute hospitalized children.

The results of this investigation show that the usage of the HC had a positive effect on pain relief in children older than 6 years undergoing venipuncture. Furthermore, the findings showed a tendency for children affected by pain and children with previous venipuncture, to a minor degree, to benefit from the HC intervention. Taking into account the difficulties of younger children in responding on scales⁴¹ and the small sample size, further research with a larger sample and on other dimensions of the acute procedural pain experience will be needed to capture the complexity of understanding and managing children's pain experience via HC intervention in different age groups. Furthermore, research will be needed pointing at psychological differences concerning the encounter with the HC in different age groups.

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